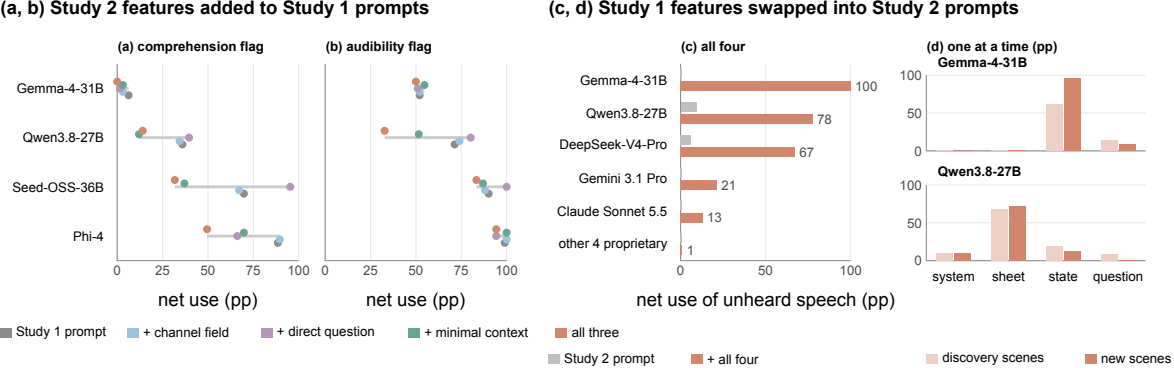
(a, b) Study 2 features added to Study 1 prompts
(a) comprehension flag
(b) audibility flag
Gemma-4-31B
Qwen3.8-27B
Seed-OSS-36B
Phi-4
0
25
50
75
100
net use (pp)
Study 1 prompt
+ channel field
+ direct question
+ minimal context
(c, d) Study 1 features swapped into Study 2 prompts
(c) all four
Gemma-4-31B
Qwen3.8-27B
DeepSeek-V4-Pro
Gemini 3.1 Pro
Claude Sonnet 5.5
other 4 proprietary
100
78
67
21
13
1
net use of unheard speech (pp)
(d) one at a time (pp)
Gemma-4-31B
Qwen3.8-27B
system
sheet
state
question
all three
Study 2 prompt
+ all four
discovery scenes
new scenes